

Samsung UN32M4500

Lab control manual — Infrared (reference), WebSocket, UPnP, and SmartThings. Not a substitute for Samsung's e-Manual.

Model	UN32M4500
Platform	18_KANTS_FHD · Tizen 3.0 · KANT-S · EDID week 4 / 2018
Name	[TV] Dave's TV television
Panel	1366×768 · 32" class
LAN	10.0.0.31 (Wi-Fi) · Samsung.local · MAC 64:1C:B0:1E:29:20
API	remote 2.0.25 · TokenAuth · WSS :8002
Developer Mode	on · Host PC 10.0.0.113 · sdb :26101
Apps (REST)	YouTube 111299001912 · Prime 3201512006785 · Disney+ 3201901017640

Open ports

Port	Role
8001	HTTP REST + WS (plain WS: unauthorized)
8002	HTTPS REST + WSS remote (pair here)
9197	UPnP DMR — SetVolume / SetMute / AVTransport
7678	AllShare UPnP
26101	sdb while Developer Mode is on
15500	Smart View / EDEN

DDC/CI on card2-DP-4 does not answer. No HDMI-CEC on FirePro mini-DP. Developer Mode is on (Host PC 10.0.0.113). Frame/Art and voice remote are off.
On disk: UN32M4500_User-manual.pdf (official 110-page e-Manual) and UN28M4500AFXZA_Service-manual.pdf (24"/28" M4500AF service book, chassis UNV72).

Protocol comparison

Infrared (first column, blue rule) is the reference: Uno 940 nm indiana-ir-send presets. **WebSocket** is the local remote stack on :8001/:8002 (keys plus REST /api/v2/). **UPnP** is AllShare DMR on :9197 (live GetVolume worked). **SmartThings** is the cloud Alexa uses; those cells are typical **2017 Tizen** capabilities, not live-called on this DUID. Green = yes · amber = partial · red = no.

Feature	Infrared reference · Uno	WebSocket :8001 / :8002	UPnP DMR :9197	SmartThings cloud / Alexa	Note
Power toggle	Partial	Yes	No	Partial	IR discrete ON/OFF; ST switch
Power on from off	Yes	Partial	No	Partial	IR POWERON; WS WOL; ST Power on with Mobile
Power off	Yes	Yes	No	Yes	IR POWEROFF
Volume step ±1	Yes	Yes	Partial	Yes	UPnP: Get then Set ±1
Set volume to N (0–100)	No	No	Yes	Yes	Beyond IR; Alexa = ST; local = UPnP
Read volume	No	No	Yes	Yes	Probed GetVolume → 14
Mute toggle	Yes	Yes	No	Partial	IR/WS MUTE
Mute on / off	Partial	No	Yes	Yes	IR unmute = VOL+
Read mute	No	No	Yes	Yes	
Source cycle	Yes	Yes	No	Yes	IR SOURCE
Set HDMI 1/2 by name	Yes	Partial	No	Yes	IR discrete HDMI1–4; WS KEY_HDMI1/2 flaky
Read current input	No	No	No	Yes	
D-pad, OK, Back, Home, Menu	Partial	Yes	No	No	IR: d-pad, OK, HOME; no Back/Menu preset
Digits 0–9	Yes	Yes	No	Partial	ST: tvChannel, not raw keys
Channel up / down / set	Partial	Yes	No	Yes	IR digits only; no CH± preset
Read channel name	No	No	No	Yes	
Launch app by ID	No	Yes	No	Partial	REST POST /applications/{id}; ST flaky on 2017
Close app	No	Yes	No	No	REST DELETE
List apps	No	Yes	No	No	samsungtv apps
App running?	No	Yes	No	Partial	REST GET /applications/{id}
Type text (IME)	Partial	Yes	No	No	IR: digits/dot only
Touchpad / mouse	No	Yes	No	No	remote_touchPad true on this set

Feature	Infrared reference · Uno	WebSocket :8001 / :8002	UPnP DMR :9197	SmartThings cloud / Alexa	Note
Media keys on TV apps	No	Partial	No	Yes	KEY_PLAY vs ST mediaPlayback
Push file/URL to TV	No	No	Yes	No	AVTransport
Control pushed media	No	No	Yes	No	
Device info	No	Yes	Partial	Yes	REST GET /api/v2/
Pairing	No	Yes	No	Partial	IR/UPnP none; WS Allow; ST account
Works TV fully off	Yes	No	No	Partial	IR POWERON
Works without internet	Yes	Yes	Yes	No	
Art Mode	No	No	No	No	FrameTVSupport: false
Change Smart Hub / ads	No	No	No	No	Settings + DNS, not APIs
Sideload apps	No	No	No	No	sdb :26101

Pick (IR as baseline): HDMI2, power on/off, vol±, Home, digits → IR when LAN is down. Absolute volume 13/33 → UPnP or SmartThings (IR cannot). Launch YouTube / type text / touchpad → samsungtv. Cast a file → UPnP. Read current input → SmartThings only.

Recipes

Developer Mode

Apps panel (not Smart Hub home) → type **12345** → On → Host PC 10.0.0.113 (or 127.0.0.1 after TizenBrew) → full power cycle.

WebSocket CLI · samsungtv

pipx samsungtvws[cli]; wrapper bin/samsungtv defaults host 10.0.0.31, port 8002, token ~/.config/indianadell/samsungtv.token. First run: Allow on the TV. Opt out SKIP_SAMSUNGTV=1. REST is not a remote: GET/POST/DELETE/PUT applications/{id} only.

```
samsungtv device-info
samsungtv home
samsungtv app-run 111299001912
samsungtv send-key KEY_SOURCE
```

UPnP absolute volume (Alexa-equivalent, local)

No pairing. POST http://10.0.0.31:9197/upnp/control/RenderingControl1 SOAPAction ...RenderingControl:1#SetVolume with <DesiredVolume>13</DesiredVolume>. GetVolume / GetMute / SetMute on the same service. Phone: YAACC (F-Droid) or BubbleUPnP — volume and cast, not HDMI/Home.

SmartThings / Alexa

“Alexa, set Dave's TV to 13” → Alexa.Speaker SetVolume → ST audioVolume.setVolume([13]). 2017: Settings → General → Network → Expert Settings → **Power On with Mobile**. Not exercised live here.

IR · indiana-ir-send

Discrete HDMI1–4 and POWER ON/OFF (NEC E0E0...). WS has no KEY_VOL11 and no reliable KEY_HDMI2. Unmute = VOL+.

Sideload / Smart Hub

Tizen 3 / Chromium 47. TizenBrew + TizenTube possible. Overscan / Moonlight WASM / Stremio Tizen 4 likely fail. APIs cannot rearrange promo tiles; use Settings ads toggles, remove TV Plus, Autorun Smart Hub off.

Not verified

SmartThings live on this DUID; KEY_HDMI2 actually switching; 32” pages in the 24/28” service SM; DDC/CI after OSD enable; WOL from full power-off.

Source: docs/samsung-un32m4500.md